P3. Assignment 3.2 Networking -Gradescope-

TCP/IP Linux Networking

Introduction

For this week's lab, you will be using **YOUR OWN LOCAL SYSTEM (the VM you created in Project 1)** to practice different Linux networking tools, all related to extracting network information and configuration. As you can imagine, computer networking is a key element in the field of cybersecurity and it is important that you know how to use these tools and their potential. To do so, this lab provides an hands-on oriented approach so that the students obtain a better understanding on how computers communicate using TCP/IP protocol.

Part 1: Fundamental networking in Linux

There are many commands in Linux that you can use to obtain information about your computer's network configuration, as well as for changing the configuration. Below is a list of commands that you'll be needing for this lab.

- ping
- *ip*
- arp
- *ss*
- hostname
- nslookup
- traceroute
- nmap
- tcpdump
- route

Description and examples of these commands can be found here:

GNU/Linux Command-Line Tools Summary (Links to an external site.).

There are other sources too. Here, you have some more:

- [13 Linux Network Configuration and Troubleshooting Commands] (https://www.tecmint.com/linux-network-configuration-and-troubleshooting-commands/ (Links to an external site.))
- 14 Useful Linux Networking Commands (Links to an external site.)

TASK 0. Record your assignment with asciinema

Start an **asciinema recording** and name it **userid.cast,** being your username the one into your NEU email (**userid**@northeastern.edu)

- (i.e.: asciinema rec useridP1.cast -t "Your Name CY5001 3.2 Assignment".

$ asciinema rec useridP1.cast -t "Your Name CY5001 3.2 Assignment" asciinema: recording asciicast to useridP1.cast asciinema: press <ctrl-d> or type "exit" when you're done

TASK 1. Find out your IP address and subnet

Use the method of your choice to find out what is your IP address, subnet and gateway.

If you use ip you will get something like this.

```
[ccl-user@localhost ~]$ ip ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
     valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
     valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
   link/ether e6:51:c7:fb:5b:c5 brd ff:ff:ff:ff:ff:ff
   inet 10.0.10.210/24 metric 100 brd 10.0.10.255 scope global dynamic eth0
     valid_lft 4873sec preferred_lft 4873sec
   inet6 fe80::e451:c7ff:fefb:5bc5/64 scope link
     valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 02:42:57:59:fb:e9 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
     valid_lft forever preferred_lft forever
```

Try to answer the following questions about TASK 1:

1. How many network interfaces do you have? List them
2. Which network interface connects you to the internet (usually through your host computer)? Provide the IP address of your VM and your subnet.
3. How many different machines (IP addresses) can potentially exist in your subnet? If you need to review this topic, you can find a good source of information here: Understand TCP/IP addressing and subnetting basics. (Links to an external site.)
4. One of the interfaces you found is labelled as LOOPBACK. Why is this interface very special? Can you think about what communications will be routed through this interface? Here (Links to an external site.) you have an explanation about loopback addresses.
5. If you needed to change your IP address to a different one in the same subnet, which command would you use?

TASK 2. Find out your gateway to the internet

The gateway of your VM is the intermediate node which routes your communication out of the VM. In Linux there is a command to access to routing configuration of your system: route. You can verify your connection with the gateway using ping

Answer the following questions on TASK 2:

1. What is your gateway IP address to connect to the internet?
2. What is the average time of an icmp connection with your gateway? Provide a screenshot

TASK 3. Hostname and other host

You can provide a name to your computer, which is known as the hostname. To do so, you can use the hostnamectl command, or edit the name in the /etc/hostname file. Also, other host names can be included in the /etc/hosts file so you can use those names instead of their IP addresses.

Answer the following questions on TASK 3:

1. What is the name of your host?
2. Change your hostname to task3host and take screenshot of the result of the output of the hostnamectl command. Make sure that the changes take effect.

3. Check which other host names are already defined in the /etc/hosts file and include a new entry third that represents the 146.190.62.39 IP address. Now take screenshot of the output of the following command ping -c 5 third

TASK 4. Check your DNS server

DNS servers are needed in order to resolve url into an ip address. If you have doubts about how DNS works, this is a good source for you to review: [What is DNS, How it Works (Links to an external site.)](#).

Answer the following questions on TASK 4:

1. Which is your configured DNS *resolver*? This is the one that will be answering your DNS queries. Take a screenshot of the output of the nslookup www.google.com command.
2. Change your DNS resolver to 8.8.8.8. Take a screenshot of the output of the nslookup www.google.com command.
3. Do you know which *transport protocol* and what *port* are used in these requests?

TASK 5. Command line Browser

Obviously, command-line internet browser is not the most fancy way to browse the internet, however for the objective of this lab, we can definitively use any of them to just trigger some communications from your Linux VM. You will need to install a command-line browser first. We recommend you to install lynx but there are many other options that you can explore here: [9 Command Line Tools for Browsing Websites and Downloading Files in Linux](#) and how to install them in your system.

Answer the following questions on TASK 5:

1. Provide a screenshot with the output of browsing http://ifconfig.me website.
2. Which application-layer protocol is used to browse the web page? Is there any other protocol involved?
3. Observe the content of the web page that you are connecting to. Notice that it displays your IP address. Is this the same IP you identified in the previous tasks of the lab? Provide some reasoning about it.

TASK 6. Capturing web traffic with tcpdump

In the previous task you were connecting to a web server ([http://ifconfig.me (Links to an external site.)](#)). Now, we want you to capture that traffic using tcpdump. To do it, you need to

open two terminals on your Linux VM, so in one you will be browsing the webpage, while in the other you will be capturing the traffic. See the example below:

Answer the following questions on TASK 6:

1. Using the command shown in the previous screenshot sudo tcpdump -A port 80 , explain each of the parameters that are used and why are they needed?
2. If you pay attention to the tcpdump capture, we browsed http://www.ifconfig.me (Links to an external site.) (you can also use http://www.google.comLinks to an external site. or http://www.northeastern.edu), but the connection is *redirected to* an https connection. Change the tcpdump command to capture both traffics (http and https). Provide your modified command in the report.
3. Now, capture HTTPS traffic going into port 443 visiting the same web servers you used before. And finally capture DNS traffic going over port 53. You should be able to trigger the corresponding traffic using different HTTPS requests.
4. **Finally, based on what you have learned about how to trigger specific type of traffic, repeat the tcpdump command with 'port 80 or port 443 or port 53' and capture the output into a PCAP file named userID.task1.6.pcap, which you need to submit it as part of your deliverables for the assignment.**

TASK 7. Network Reconnaissance

Network Mapper, nmap is a Linux command line tool for network reconnaissance and security auditing. It can be use by security administrators to audit what network services are open in a specific subnet and, at the same time, it is a tool commonly used by hackers as a reconnaissance tool to discover vulnerable network services. Although nmap is command line tool, it can also provide very rich information about the services, operating systems, …

There are an important number of parameters to tweak nmap behaviour, you can read most important of them here (Links to an external site.). Most parameters are intended for security administrators, but others have more controversial purposes like decoying or hide the scan from victim's sight, such as -sS stealth mode, sA detect firewall or D decoy origin. Those last commands will require root privileges to execute.

Here you have an example of how to run a fast nmap search over a subnet

nmap -F 172.17.0.*

Starting Nmap 6.40 (http://nmap.org) at 2020-10-09 11:52 UTC

Nmap scan report for 172.17.0.1

Host is up (0.00038s latency).

Not shown: 98 closed ports

PORT STATE SERVICE

22/tcp open ssh

111/tcp open rpcbind

Nmap done: 256 IP addresses (1 host up) scanned in 3.12 seconds

Answer the following questions on TASK 7:

1. Provide a screenshot with the output of scanning the subnet where your VM is attached to.
2. Select one machine in your subnetwork and perform a nmap scan which provides information about the operating system of the target machine. Provide a screenshot of the command and the output.

You may need to provide root privileges using sudo.

3. Execute an Stealth Mode search with nmap and take a screenshot of the output.

4. Explain how the Stealth Mode works, and how it's more OpSec friendly than other scanning modes ?

Part 2: Fundamental networking in Docker containers

TASK 0. Record your assignment with asciinema

Start an **asciinema recording** and name it **userid.cast,** being your username the one into your NEU email (**userid**@northeastern.edu)

- (i.e.: asciinema rec useridP2.cast -t "Your Name CY5001 3.2 Assignment".

$ asciinema rec useridP2.cast -t "Your Name CY5001 3.2 Assignment"

asciinema: recording asciicast to useridP2.cast

asciinema: press <ctrl-d> or type "exit" when you're done

TASK 1. Understanding Bridge Networks

Create a custom bridge network named cy5001net and verify that it is correctly created:

```
$ docker network create cy5001net
$ docker network ls
NETWORK ID     NAME        DRIVER    SCOPE
84855dd0f597   bridge      bridge    local
70b8a77f8e69   cy5001net   bridge    local
858cf1562c99   host        host      local
1adb23f88c22   none        null      local
```

Answer the following questions on TASK 1:

1. When creating a custom bridge network like cy5001net, how does Docker's network isolation enhance the security of containers connected to this network compared to containers connected to the default bridge network?
2. What are the potential security risks of using a default bridge network for containers, and how can using a custom bridge network help mitigate these risks?

TASK 2. Running containers on the cy5001net

Run two containers and connect them to cy5001net and verify that both are running and can communicate.

```
$ docker run -dit --name containerA --network cy5001net alpine sh
$ docker run -dit --name containerB --network cy5001net alpine sh

$  docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS         PORTS     NAMES
314919df4852   alpine    "sh"      10 minutes ago   Up 10 minutes            containerB
5ed48ec0c39f   alpine    "sh"      10 minutes ago   Up 10 minutes            containerA

$ docker exec -it containerB ping containerA
PING containerA (172.18.0.5): 56 data bytes
64 bytes from 172.18.0.5: seq=0 ttl=64 time=0.127 ms
64 bytes from 172.18.0.5: seq=1 ttl=64 time=0.131 ms
...
```

```
$ docker exec -it containerA ping containerB
PING containerB (172.18.0.6): 56 data bytes
64 bytes from 172.18.0.6: seq=0 ttl=64 time=0.071 ms
64 bytes from 172.18.0.6: seq=1 ttl=64 time=0.080 ms
...
```

Now you can clean up and remove the containers and the bridge network you created in this task.

Answer the following questions on TASK 2:

1. Given that containers on the same bridge network, such as cy5001net, can communicate freely with each other, what security concerns might arise if a container on this network is compromised? How could this affect other containers on the same network?
2. How would you secure communication between containers on the cy5001net network to prevent unauthorized access or interception of data? Discuss any tools you could use.

TASK 3. Running containers on *host* network stack

Run a new container on the *host* network and install a httpd and ssh server on it.

```
$ docker run -dit --name containerC --network host ubuntu sh
$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS         PORTS     NAMES
09c02a624b95   alpine    "sh"      4 minutes ago   Up 4 minutes             containerC
```

Now *containerC* is using the same network stack than the host, so because httpd opens port 80 by default for http connections you should be able to connect to that web server using the same ip of the host machine.

```
$ netstat -ntlp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address          Foreign Address        State       PID/Program name
tcp        0      0 127.0.0.1:36969        0.0.0.0:*              LISTEN      -
tcp        0      0 127.0.0.53:53          0.0.0.0:*              LISTEN      -
tcp        0      0 0.0.0.0:22             0.0.0.0:*              LISTEN      -
tcp6       0      0 :::80                  :::*                   LISTEN      -
```

```
tcp6       0      0 :::22                   :::*                    LISTEN      -
instructor@cy5001:~$ curl localhost
<html><body><h1>It works!</h1></body></html>
```

Keep in mind that this happens without having to map any ports because the container is using the same network stack than the host (basically the same network interface). Also, if your host system has a firewall active blocking server ports < 1024, you may need to edit the /usr/local/apache2/conf/httpd.conf file in the container and start the httpd service in an upper port (e.g. 8080).

Answer the following questions on TASK 3:

1. Using the host network mode allows a container to share the host's network stack. What are the potential security implications of this configuration, especially if the container is compromised?
2. In a production environment, why might you avoid using the host network mode from a security perspective? What alternative Docker network modes could provide better isolation?

TASK 4. Running container without network

Run a container with the network mode set to none and then check container's connectivity.

```
$ docker run -dit --name containerD --network none alpine sh
d87f5868ebb1a7c326fbe30f299064bde826ef76be5ff08f100548cc61a7e95a

$ docker exec -it containerD ping google.com
ping: bad address 'google.com'
```

Answer the following questions on TASK 4:

1. Running a container with the network mode set to none isolates it from all networks. How does this configuration enhance the security of the container, and in what scenarios might this be particularly useful?
2. Despite the enhanced security provided by the none network mode, what are the limitations of this approach, and how could they impact the container's ability to perform its intended functions?

Deliverables

Deliverable for this assignment is submitted via Gradescope, submission file names need to be exact (if not points will be deducted):

- Your asciinema recording from each part useridP1.cast and useridP2.cast. We recommend that you verify that your cast files can be played with asciinema before submitting them.
- A PDF document report with screenshots and answers to the questions of each task. You need to reference your answers and screenshots, so it is clear to what PART and TASK you are referring to (e.g TASK 1.7 for PART 1, TASK 7).
- The traffic captured file userid.task1.6.pcap you created in part 1, task 6. We recommend you to verify that your capture can be read in your system before it is submitted (with tcpdump -r).

Grading

Please see here the rubric for the assignment.

Here you have some more interesting sources for reference:

https://danielmiessler.com/study/tcpdump/Links to an external site.

https://www.tecmint.com/command-line-web-browsers/Links to an external site.